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पो.ब.नं. : ७४३७

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**DEPARTMENT OF COMPUTER SCIENCE****COMPUTER NETWORK****LAB 2****Windows Basic Networking Commands and DHCP Network Configuration**

Submitted By:	Submitted To:	Signature:
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Lab No. 2

Title: Windows Networking Utilities and DHCP Network Configuration

Lab 2.1: Introduction to Windows Network Administration Commands

Objective: To gain hands-on experience with Windows CLI tools utilized in network management, troubleshooting, and configuration.

Theory

Microsoft Windows includes a powerful array of built-in networking command-line tools tailored for network administration, configuration, and diagnostics. These utilities allow administrators to effortlessly audit IP configurations, test end-to-end network connectivity, and troubleshoot domain name resolution issues. Additionally, they offer critical insights into active data paths and real-time network connections, making it easier to monitor traffic, isolate performance issues, and efficiently manage network services.

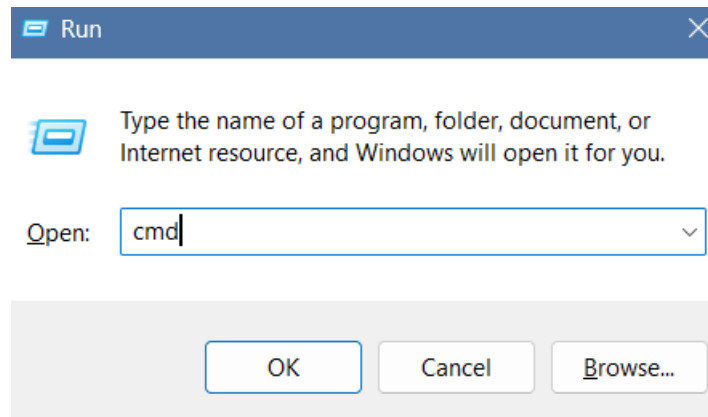
Components Required

1. Computer System
2. Windows Operating System
3. Command Prompt (CMD)
4. Network Connection

Procedure:

1. Opening Command Prompt

1. Press Windows + R.
2. Type cmd or command prompt and press Enter.
3. The Command Prompt window will open (without administration access at first)



Key Commands used:

1. ipconfig

Provides a quick overview of the host's primary network settings: IP address, subnet mask, and default gateway.

Syntax: ipconfig

```
C:\Users\user>ipconfig

Windows IP Configuration

Ethernet adapter vEthernet (Default Switch):

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::bb02:9ad8:cdc1:6515%23
    IPv4 Address. . . . . : 172.27.80.1
    Subnet Mask . . . . . : 255.255.240.0
    Default Gateway . . . . . : 

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::3d33:efd6:b4fe:5e00%16
    IPv4 Address. . . . . : 10.0.100.36
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 10.0.100.1

Wireless LAN adapter Local Area Connection* 9:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Local Area Connection* 10:
```

```
Connection-specific DNS Suffix . :  
Link-local IPv6 Address . . . . . : fe80::b3ad:e52c:47c7:d097%20  
IPv4 Address. . . . . : 192.168.137.1  
Subnet Mask . . . . . : 255.255.255.0  
Default Gateway . . . . . :
```

Wireless LAN adapter Wi-Fi:

```
Media State . . . . . : Media disconnected  
Connection-specific DNS Suffix . :
```

Ethernet adapter Bluetooth Network Connection 2:

```
Media State . . . . . : Media disconnected  
Connection-specific DNS Suffix . :
```

2. ipconfig /all

Provides an exhaustive breakdown of all network adapters, including MAC addresses, DNS servers, and DHCP lease details.

Syntax: ipconfig /all

```
C:\Users\user>ipconfig/all  
  
Windows IP Configuration  
  
Host Name . . . . . : DESKTOP-MF454V4  
Primary Dns Suffix . . . . . :  
Node Type . . . . . : Mixed  
IP Routing Enabled. . . . . : No  
WINS Proxy Enabled. . . . . : No  
  
Ethernet adapter vEthernet (Default Switch):  
  
Connection-specific DNS Suffix . :  
Description . . . . . : Hyper-V Virtual Ethernet Adapter  
Physical Address. . . . . : 00-15-5D-07-29-05  
DHCP Enabled. . . . . : No  
Autoconfiguration Enabled . . . . : Yes  
Link-local IPv6 Address . . . . . : fe80::bb02:9ad8:cdc1:6515%23(Preferred)  
IPv4 Address. . . . . : 172.27.80.1(Preferred)  
Subnet Mask . . . . . : 255.255.240.0  
Default Gateway . . . . . :  
DHCPv6 IAID . . . . . : 385881437  
DHCPv6 Client DUID. . . . . : 00-01-00-01-2E-FB-B4-09-40-C2-BA-FE-9C-28  
NetBIOS over Tcpip. . . . . : Enabled  
  
Ethernet adapter Ethernet:  
  
Connection-specific DNS Suffix . :  
Description . . . . . : Realtek PCIe GbE Family Controller  
Physical Address. . . . . : 40-C2-BA-FE-9C-28  
DHCP Enabled. . . . . : Yes  
Autoconfiguration Enabled . . . . : Yes  
Link-local IPv6 Address . . . . . : fe80::3d33:efd6:b4fe:5e00%16(Preferred)  
IPv4 Address. . . . . : 10.0.100.36(Preferred)  
Subnet Mask . . . . . : 255.255.255.0  
Lease Obtained. . . . . : Wednesday, June 24, 2026 9:36:43 AM  
Lease Expires . . . . . : Wednesday, June 24, 2026 4:16:42 PM  
Default Gateway . . . . . : 10.0.100.1  
DHCP Server . . . . . : 10.0.100.1  
DHCPv6 IAID . . . . . : 88130234  
DHCPv6 Client DUID. . . . . : 00-01-00-01-2E-FB-B4-09-40-C2-BA-FE-9C-28  
DNS Servers . . . . . : 8.8.8.8  
                              8.8.4.4  
NetBIOS over Tcpip. . . . . : Enabled  
  
Wireless LAN adapter Local Area Connection* 9:
```

3. ipconfig /release

Releases the current IP address obtained from the DHCP server.

Syntax: ipconfig /release

```
C:\Users\user>ipconfig/release

Windows IP Configuration

No operation can be performed on Local Area Connection* 9 while it has its media disconnected.
No operation can be performed on Wi-Fi while it has its media disconnected.
No operation can be performed on Bluetooth Network Connection 2 while it has its media disconnected.

Ethernet adapter vEthernet (Default Switch):

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::bb02:9ad8:cdc1:6515%23
    IPv4 Address. . . . . : 172.27.80.1
    Subnet Mask . . . . . : 255.255.240.0
    Default Gateway . . . . . : 

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::3d33:efd6:b4fe:5e00%16
    Default Gateway . . . . . : 

Wireless LAN adapter Local Area Connection* 9:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Local Area Connection* 10:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::b3ad:e52c:47c7:d097%20
    IPv4 Address. . . . . : 192.168.137.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 

Wireless LAN adapter Wi-Fi:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Ethernet adapter Bluetooth Network Connection 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :
```

4. ipconfig /renew

Requests a new IP address from the DHCP server.

Syntax: ipconfig /renew

```
C:\WINDOWS\system32\cmd. X + -
C:\Users\user>ipconfig/renew

Windows IP Configuration

No operation can be performed on Local Area Connection* 9 while it has its media disconnected.
No operation can be performed on Wi-Fi while it has its media disconnected.
No operation can be performed on Bluetooth Network Connection 2 while it has its media disconnected.

Ethernet adapter vEthernet (Default Switch):

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::9d90:6eca:5047:a18a%23
    IPv4 Address. . . . . : 172.27.80.1
    Subnet Mask . . . . . : 255.255.240.0
    Default Gateway . . . . . : 

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::3d33:efd6:b4fe:5e00%16
    IPv4 Address. . . . . : 10.0.100.36
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 10.0.100.1

Wireless LAN adapter Local Area Connection* 9:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Local Area Connection* 10:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::b3ad:e52c:47c7:d097%20
    IPv4 Address. . . . . : 192.168.137.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 

Wireless LAN adapter Wi-Fi:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Ethernet adapter Bluetooth Network Connection 2:

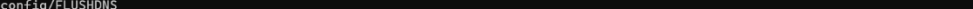
    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

C:\Users\user>
```

5. ipconfig /flushdns

Clears the DNS resolver cache stored on the local computer.

Syntax: `ipconfig /flushdns`



```
C:\Users\user>ipconfig /flushdns
Windows IP Configuration
Successfully flushed the DNS Resolver Cache.
C:\Users\user>
```

6.ping

Tests connectivity between two devices by sending ICMP Echo Request packets.

Syntax: ping <IP Address> or ping website URL.

Example: ping 8.8.8.8 or ping sxc.edu.np

```
C:\Users\user>ping sxc.edu.np

Pinging sxc.edu.np [202.51.0.118] with 32 bytes of data:
Reply from 202.51.0.118: bytes=32 time=4ms TTL=57
Reply from 202.51.0.118: bytes=32 time=3ms TTL=57
Reply from 202.51.0.118: bytes=32 time=3ms TTL=57
Reply from 202.51.0.118: bytes=32 time=3ms TTL=57

Ping statistics for 202.51.0.118:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 4ms, Average = 3ms

C:\Users\user>
```

7.ping -t

Continuously sends ping requests until manually stopped using Ctrl + C.

Syntax: ping <Address> -t

Example: `ping google.com -t`

[illegible]

8. ping -l <size> -t

Sends continuous ping requests with a specified packet size.

Syntax: ping <IP Address> -l <packet_size> -t

Example: ping google.com -l 1500 -t

```
C:\Users\user>ping google.com -l 1500 -t

Pinging google.com [142.251.220.110] with 1500 bytes of data:
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 142.251.220.110:
    Packets: Sent = 3, Received = 0, Lost = 3 (100% loss),
    Control-C
^C
C:\Users\user>
```

9. tracert

Displays the path taken by packets from the source to the destination.

Syntax: tracert <Address>

Example: i. tracert google.com

ii. tracert sxc.edu.np

```
C:\Users\user>tracert google.com

Tracing route to google.com [142.250.183.174]
over a maximum of 30 hops:
  0  <1 ms  <1 ms  <1 ms  10.0.100.1
  1  1 ms  1 ms  <1 ms  103.41.172.11
  2  2 ms  8 ms  1 ms  100.127.192.45
  3  1 ms  1 ms  1 ms  43.245.84.210
  4  1 ms  1 ms  1 ms  ae0-bq2.vianet.com.np [110.44.112.66]
  5  71 ms  44 ms  44 ms  125.18.98.249
  6  40 ms  *  40 ms  hosteurope-ic-371682.ip.twelve99-cust.net [213.248.77.119]
  7  42 ms  42 ms  43 ms  74.125.147.238
  8  40 ms  40 ms  40 ms  209.85.251.159
  9  40 ms  40 ms  41 ms  142.251.55.65
 10  41 ms  40 ms  40 ms  lcmaaa-ar-in-f14.1e100.net [142.250.183.174]

Trace complete.

C:\Users\user>
```

fig I. tracert google.com

```
C:\Users\user>tracert sxc.edu.np

Tracing route to sxc.edu.np [202.51.0.118]
over a maximum of 30 hops:
  0  <1 ms  <1 ms  <1 ms  10.0.100.1
  1  1 ms  1 ms  1 ms  103.41.172.11
  2  1 ms  1 ms  1 ms  100.127.192.45
  3  1 ms  1 ms  1 ms  103.10.28.10
  4  1 ms  1 ms  1 ms  198.32.231.168
  5  *  *  *  Request timed out.
  6  *  *  *  Request timed out.
  7  3 ms  2 ms  2 ms  202.51.0.118

Trace complete.

C:\Users\user>
```

Fig II. Tracert sxc.edu.np

10. nslookup

Queries DNS servers to obtain IP addresses corresponding to domain names.

Syntax: nslookup <domain_name>

Example: nslookup google.com

```
C:\Users\user>nslookup
Default Server: dns.google
Address: 8.8.8.8

> sxc.edu.np
Server: dns.google
Address: 8.8.8.8

Non-authoritative answer:
Name: sxc.edu.np
Address: 202.51.0.118

> daraz.np
Server: dns.google
Address: 8.8.8.8

*** dns.google can't find daraz.np: Non-existent domain
> daraz.com.np
Server: dns.google
Address: 8.8.8.8

Non-authoritative answer:
Name: daraz.com.np
Addresses: 47.246.174.23
           47.246.167.168
           47.246.167.152
           47.246.174.148

> onlinekhabar.com
Server: dns.google
Address: 8.8.8.8

Non-authoritative answer:
Name: onlinekhabar.com
Addresses: 2606:4700:20::681a:492
           2606:4700:20::681a:592
           2606:4700:20::ac43:470e
           104.26.5.146
           104.26.4.146
           172.67.71.14

>
```

11. netstat

Displays active network connections and network statistics.

Syntax: netstat

```
C:\WINDOWS\system32\cmd. x
Active Connections

Proto Local Address Foreign Address State
TCP 10.0.100.36:49537 4.213.25.241:https ESTABLISHED
TCP 10.0.100.36:49668 si-in-f188:5228 ESTABLISHED
TCP 10.0.100.36:51197 lb-140-82-114-26-iad:https ESTABLISHED
TCP 10.0.100.36:51940 160.79.104.10:https ESTABLISHED
TCP 10.0.100.36:52572 104.18.32.47:https ESTABLISHED
TCP 10.0.100.36:53087 172.64.148.235:https ESTABLISHED
TCP 10.0.100.36:54404 pnmaaa-ay-in-f3:https ESTABLISHED
TCP 10.0.100.36:55891 pnmaaa-av-in-f10:https ESTABLISHED
TCP 10.0.100.36:60004 maa05s20-in-f10:https TIME_WAIT
TCP 10.0.100.36:61019 20.192.44.82:https ESTABLISHED
TCP 10.0.100.36:61849 dns:https ESTABLISHED
TCP 10.0.100.36:62090 151.101.66.137:https ESTABLISHED
TCP 10.0.100.36:62559 pnmaaa-ay-in-f3:https ESTABLISHED
TCP 10.0.100.36:64863 lcomaa-am-in-f10:https TIME_WAIT
TCP 10.0.100.36:64964 pnmaaa-bf-in-f10:https ESTABLISHED
TCP 10.0.100.36:64966 server-18-239-153-85:https ESTABLISHED
TCP 127.0.0.1:5141 checkhost:65294 ESTABLISHED
TCP 127.0.0.1:46933 checkhost:53407 ESTABLISHED
TCP 127.0.0.1:49671 checkhost:49672 ESTABLISHED
TCP 127.0.0.1:49672 checkhost:49671 ESTABLISHED
TCP 127.0.0.1:49679 checkhost:49680 ESTABLISHED
TCP 127.0.0.1:49680 checkhost:49679 ESTABLISHED
TCP 127.0.0.1:49985 checkhost:49986 ESTABLISHED
TCP 127.0.0.1:49986 checkhost:49985 ESTABLISHED
TCP 127.0.0.1:51779 checkhost:64762 ESTABLISHED
TCP 127.0.0.1:53407 checkhost:46933 ESTABLISHED
TCP 127.0.0.1:58995 checkhost:61495 ESTABLISHED
TCP 127.0.0.1:61495 checkhost:58995 ESTABLISHED
TCP 127.0.0.1:62911 checkhost:62912 ESTABLISHED
TCP 127.0.0.1:62912 checkhost:62911 ESTABLISHED
TCP 127.0.0.1:62913 checkhost:62914 ESTABLISHED
TCP 127.0.0.1:62914 checkhost:62913 ESTABLISHED
TCP 127.0.0.1:64762 checkhost:51779 ESTABLISHED
TCP 127.0.0.1:65294 checkhost:5141 ESTABLISHED
TCP [::1]:4343 DESKTOP-MF454V4:59214 ESTABLISHED
TCP [::1]:4449 DESKTOP-MF454V4:59216 ESTABLISHED
TCP [::1]:5141 DESKTOP-MF454V4:59215 ESTABLISHED
TCP [::1]:15150 DESKTOP-MF454V4:56255 ESTABLISHED
TCP [::1]:15150 DESKTOP-MF454V4:59213 ESTABLISHED
TCP [::1]:56255 DESKTOP-MF454V4:15150 ESTABLISHED
TCP [::1]:59213 DESKTOP-MF454V4:15150 ESTABLISHED
TCP [::1]:59214 DESKTOP-MF454V4:4343 ESTABLISHED
TCP [::1]:59215 DESKTOP-MF454V4:5141 ESTABLISHED
TCP [::1]:59216 DESKTOP-MF454V4:4449 ESTABLISHED

C:\Users\user>
```


12. netstat -an

Displays all active connections and listening ports numerically.

Syntax: netstat -an

```
C:\WINDOWS\system32\cmd. x + v
UDP 0.0.0.0:61636 *:*
UDP 10.0.100.36:137 *:*
UDP 10.0.100.36:138 *:*
UDP 10.0.100.36:1900 *:*
UDP 10.0.100.36:62490 *:*
UDP 127.0.0.1:1900 *:*
UDP 127.0.0.1:49547 *:*
UDP 127.0.0.1:50510 127.0.0.1:50510
UDP 127.0.0.1:62466 127.0.0.1:62466
UDP 127.0.0.1:62492 *:*
UDP 172.27.80.1:67 *:*
UDP 172.27.80.1:68 *:*
UDP 172.27.80.1:137 *:*
UDP 172.27.80.1:138 *:*
UDP 172.27.80.1:1900 *:*
UDP 172.27.80.1:62489 *:*
UDP 192.168.137.1:67 *:*
UDP 192.168.137.1:68 *:*
UDP 192.168.137.1:137 *:*
UDP 192.168.137.1:138 *:*
UDP 192.168.137.1:1900 *:*
UDP 192.168.137.1:62491 *:*
UDP [::]:547 *:*
UDP [::]:5353 *:*
UDP [::]:5353 *:*
UDP [::]:5353 *:*
UDP [::]:5353 *:*
UDP [::]:5353 *:*
UDP [::]:5353 *:*
UDP [::]:5355 *:*
UDP [::]:49154 *:*
UDP [::]:49553 *:*
UDP [::]:49555 *:*
UDP [::]:49565 *:*
UDP [::]:53066 *:*
UDP [::]:53247 *:*
UDP [::]:61636 *:*
UDP [::1]:1900 *:*
UDP [::1]:62488 *:*
UDP [fe80::3d33:efd6:b4fe:5e00%16]:1900 *:*
UDP [fe80::3d33:efd6:b4fe:5e00%16]:49665 *:*
UDP [fe80::9d90:6eca:5047:a18a%23]:1900 *:*
UDP [fe80::9d90:6eca:5047:a18a%23]:59482 *:*
UDP [fe80::b3ad:e52c:47c7:d097%20]:53 *:*
UDP [fe80::b3ad:e52c:47c7:d097%20]:1900 *:*
UDP [fe80::b3ad:e52c:47c7:d097%20]:62487 *:*
C:\Users\user>
```

13. arp -a

Displays the Address Resolution Protocol (ARP) cache containing IP-to-MAC address mappings.

Syntax: arp -a

```
C:\Users\user>arp -a
Interface: 10.0.100.36 --- 0x10
Internet Address Physical Address Type
10.0.100.1 c8-4f-86-fc-00-05 dynamic
10.0.100.255 ff-ff-ff-ff-ff-ff static
224.0.0.22 01-00-5e-00-00-16 static
224.0.0.251 01-00-5e-00-00-fb static
224.0.0.252 01-00-5e-00-00-fc static
239.255.255.250 01-00-5e-7f-ff-fa static
255.255.255.255 ff-ff-ff-ff-ff-ff static

Interface: 192.168.137.1 --- 0x14
Internet Address Physical Address Type
192.168.137.29 de-38-31-4e-26-97 static
192.168.137.32 2a-08-dd-73-3b-4c static
192.168.137.58 f2-29-4a-a7-e2-68 static
192.168.137.149 2a-61-e3-b0-c0-f9 static
192.168.137.164 a0-02-a5-cb-52-dc static
192.168.137.174 14-d4-24-54-5a-39 static
192.168.137.197 ae-76-67-16-9c-a2 static
192.168.137.255 ff-ff-ff-ff-ff-ff static
224.0.0.2 01-00-5e-00-00-02 static
224.0.0.22 01-00-5e-00-00-16 static
224.0.0.251 01-00-5e-00-00-fb static
224.0.0.252 01-00-5e-00-00-fc static
224.0.0.253 01-00-5e-00-00-fd static
239.255.255.250 01-00-5e-7f-ff-fa static
255.255.255.255 ff-ff-ff-ff-ff-ff static

Interface: 172.27.80.1 --- 0x17
Internet Address Physical Address Type
172.27.95.255 ff-ff-ff-ff-ff-ff static
224.0.0.22 01-00-5e-00-00-16 static
224.0.0.251 01-00-5e-00-00-fb static
239.255.255.250 01-00-5e-7f-ff-fa static
255.255.255.255 ff-ff-ff-ff-ff-ff static
C:\Users\user>
```

14. hostname

provides the name of the local computer or the user in the network.

Syntax: hostname

```
C:\Users\user>hostname
DESKTOP-MF454V4
C:\Users\user>|
```



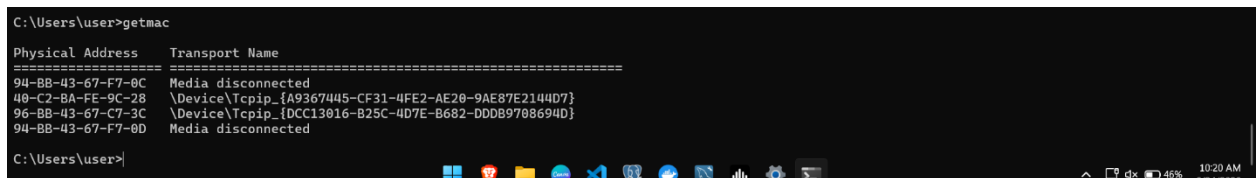
15. getmac

Displays the MAC addresses of network adapters installed on the computer.

Syntax: getmac

```
C:\Users\user>getmac

Physical Address   Transport Name
-----
94-BB-43-67-F7-8C Media disconnected
48-C2-BA-FE-9C-28 \Device\NPF{A9367445-CF31-4FE2-AE28-9AE87E2144D7}
96-BB-43-67-C7-3C \Device\NPF{DCC13016-B25C-4D7E-B682-DDDB9708694D}
94-BB-43-67-F7-8D Media disconnected
C:\Users\user>|
```



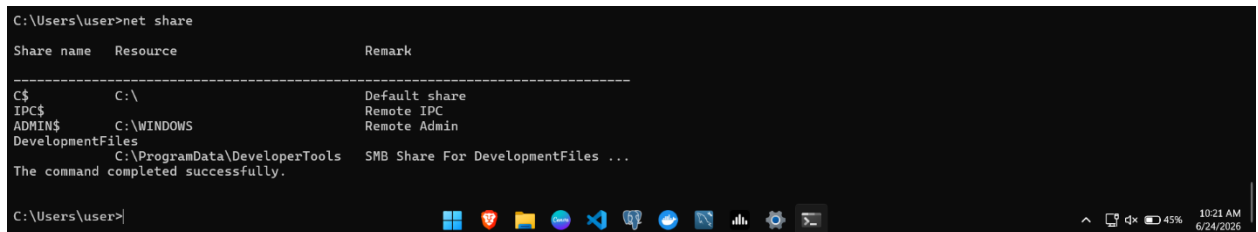
16. net share

Displays shared folders and resources available on the computer.

Syntax: net share

```
C:\Users\user>net share

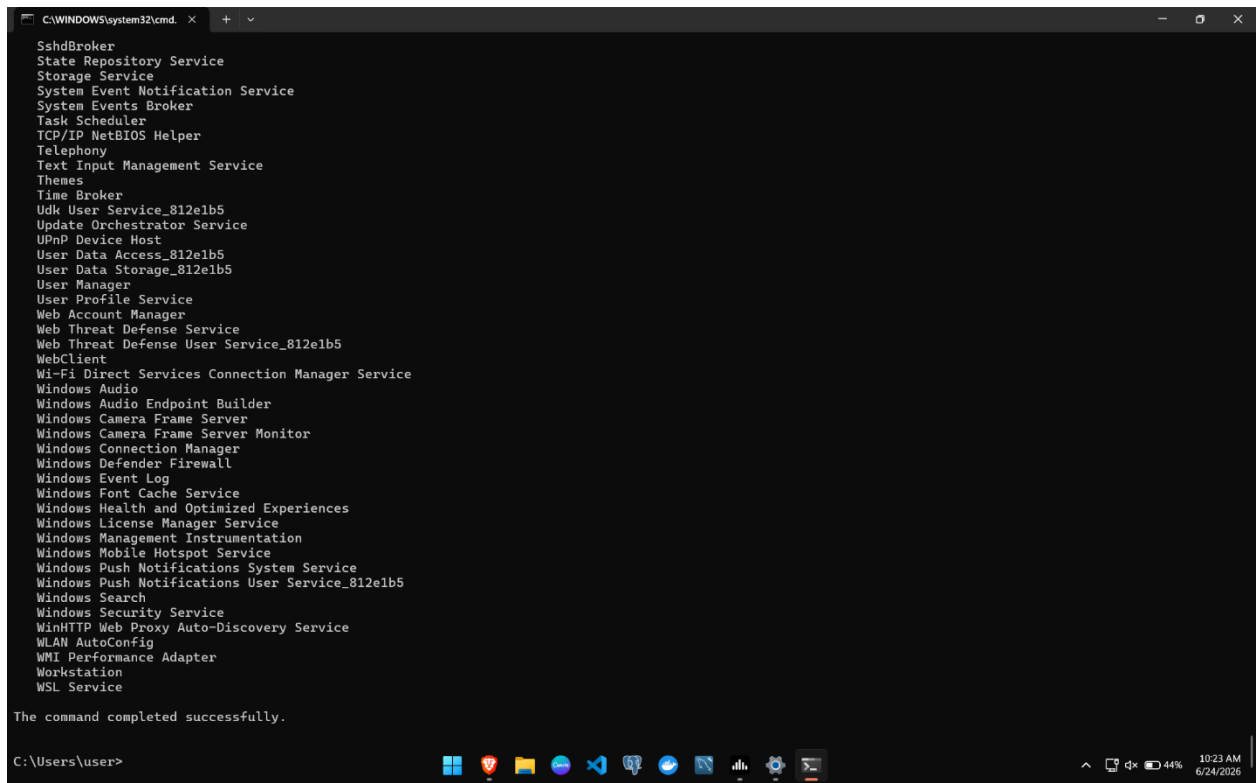
Share name   Resource                                Remark
-----
C$           C:\                                    Default share
IPC$         C:\                                   Remote IPC
ADMIN$       C:\WINDOWS                           Remote Admin
DevelopmentFiles
             C:\ProgramData\DeveloperTools       SMB Share For DevelopmentFiles ...
The command completed successfully.
C:\Users\user>|
```



17. net start

Displays currently running Windows services.

Syntax: net start



```
C:\WINDOWS\system32\cmd. x + -
SchdBroker
State Repository Service
Storage Service
System Event Notification Service
System Events Broker
Task Scheduler
TCP/IP NetBIOS Helper
Telephony
Text Input Management Service
Themes
Time Broker
Udk User Service_812e1b5
Update Orchestrator Service
UPnP Device Host
User Data Access_812e1b5
User Data Storage_812e1b5
User Manager
User Profile Service
Web Account Manager
Web Threat Defense Service
Web Threat Defense User Service_812e1b5
WebClient
Wi-Fi Direct Services Connection Manager Service
Windows Audio
Windows Audio Endpoint Builder
Windows Camera Frame Server
Windows Camera Frame Server Monitor
Windows Connection Manager
Windows Defender Firewall
Windows Event Log
Windows Font Cache Service
Windows Health and Optimized Experiences
Windows License Manager Service
Windows Management Instrumentation
Windows Mobile Hotspot Service
Windows Push Notifications System Service
Windows Push Notifications User Service_812e1b5
Windows Search
Windows Security Service
WinHTTP Web Proxy Auto-Discovery Service
WLAN AutoConfig
WMI Performance Adapter
Workstation
WSL Service

The command completed successfully.

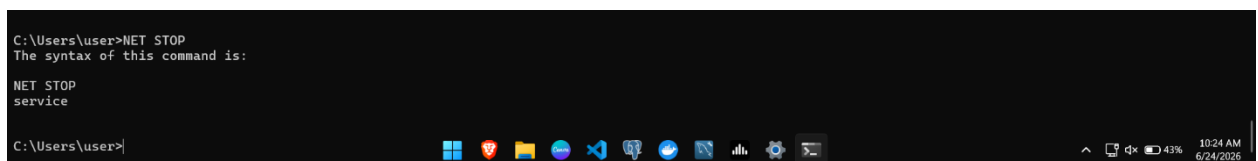
C:\Users\user>
```

18. net stop

Stops a specified Windows service.

Syntax: net stop <service_name>

Example: net stop spooler



```
C:\Users\user>NET STOP
The syntax of this command is:

NET STOP
service

C:\Users\user>
```

19. color

Changes the foreground and background colors of the Command Prompt window.

Syntax: color n (where n is the integer)

```
C:\Users\User>color 2
C:\Users\User>color 1
Sets the default console foreground and background colors.

COLOR [attr]

    attr            Specifies color attribute of console output

Color attributes are specified by TWO hex digits -- the first
corresponds to the background; the second the foreground. Each digit
can be any of the following values:

    0 = Black      8 = Gray
    1 = Blue       9 = Light Blue
    2 = Green      A = Light Green
    3 = Aqua       B = Light Aqua
    4 = Red        C = Light Red
    5 = Purple     D = Light Purple
    6 = Yellow     E = Light Yellow
    7 = White      F = Bright White

If no argument is given, this command restores the color to what it was
when CMD.EXE started. This value either comes from the current console
window, the /I command line switch or from the DefaultColor registry
value.

The COLOR command sets ERRORLEVEL to 1 if an attempt is made to execute
the COLOR command with a foreground and background color that are the
same.

Example: "COLOR fc" produces light red on bright white

C:\Users\User>
```

Examples: color 1F: Blue background with bright white text.

color 2F: Green background with bright white text.

color 3F: Aqua background with bright white text.

Lab 2.2: DHCP Server Configuration and Dynamic IP Address Allocation Using Cisco Packet Tracer

Objective: Demonstrates how to configure a DHCP server and validate automatic IP address deployment to network hosts within Packet Tracer.

Theory

While Lab 1 utilized manual (static) IP addressing, a method viable only for small-scale environments. Scaling a network makes manual configuration highly inefficient and prone to human error.

To resolve this management bottleneck, Dynamic Host Configuration Protocol (DHCP) is deployed. Operating at the application layer, DHCP automates the distribution of IP addresses alongside critical network parameters like subnet masks, default gateways, and DNS server addresses. This automation significantly reduces administrative overhead and eliminates configuration conflicts across the network.

COMPONENTS IN THE DHCP CONFIGURATION

1. DHCP SERVER

A DHCP server manages a centralized IP address pool, dynamically allocating addresses to network clients upon request.

2. DHCP Client

A DHCP client is any network host configured to automatically request and receive its IP parameters from a DHCP server.

3. DHCP Scope

The range of IP addresses available for assignment.

Example:

Network: 192.168.1.0/24

Range: 192.168.1.2 – 192.168.1.254

Gateway: 192.168.1.1

DHCP Process (DORA)

I. DISCOVER

The client broadcasts a DHCP Discover message to locate available DHCP servers.

II. OFFER

The DHCP server replies with an available IP address.

III. REQUEST

The client requests the offered IP address.

IV. ACKNOWLEDGEMENT

This final confirmation completes the DORA transaction—a four-step message exchange consisting of Discover, Offer, Request, and Acknowledgment—which governs how a client secures an IP lease.

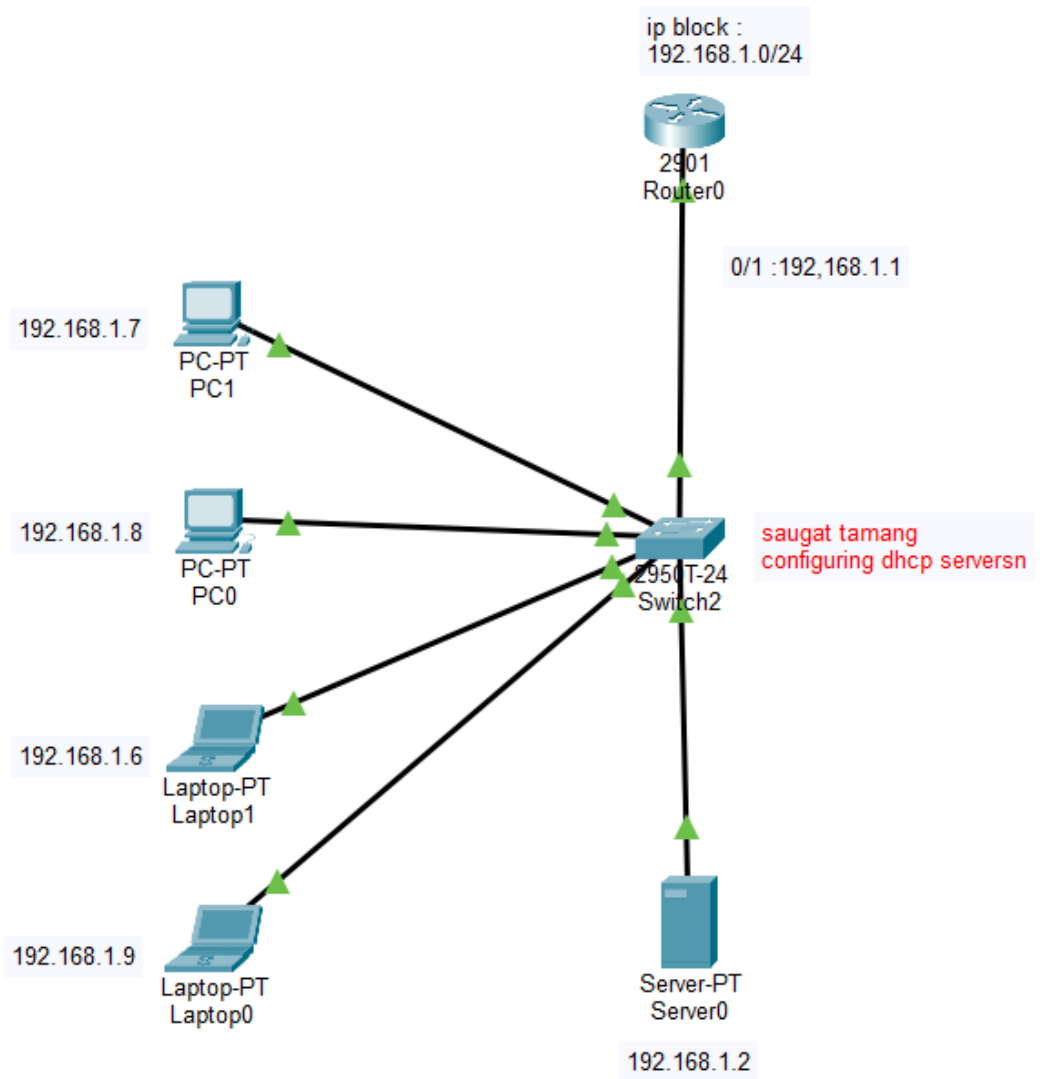
Components Required

1. Router 2901
2. Switch 2960
3. Server
4. PCs/Laptops
5. Copper Straight-Through Cable

Procedure:

1. Creating the Network Topology

1. Launch Cisco Packet Tracer and open a blank workspace.
2. Place the following devices in the workspace:
 - 1 * Router (2901)
 - 1* Switch (2960)
 - 1* Server
 - 2 * PCs
 - 2 * Laptops
3. Interconnect all host endpoints and the server to the central Switch using standard Copper Straight-Through media.
4. Connect the Router's LAN interface to an available port on the Switch using a Copper Straight-Through cable to finalize physical topology.



2. Configuring the Router

1. Open the Router CLI.
2. Enter privileged mode: *Router> enable*
3. Enter configuration mode: *Router# conf t*
4. Configure the router interface:

```
Router(config)# interface gigabitEthernet 0/0
Router(config-if)# ip address 192.168.1.1 255.255.255.0
Router(config-if)# no shutdown
```

5. Save the configuration.

 Router0

Physical Config CLI Attributes

IOS Command Line Interface

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gigabitEthernet 0/1
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
```


3. Configuring DHCP on the Router

1. Open the Router CLI and enter global configuration mode.

```
Router> enable
```

```
Router# configure terminal
```

2. Exclude reserved IP addresses:

```
Router(config)# ip dhcp excluded-address 192.168.1.1 192.168.1.5
```

3. Create a DHCP pool: *Router(config)# ip dhcp pool surakshya_dahal*

4. Configure the network, gateway, and DNS server:

```
Router(dhcp-config)# network 192.168.1.0 255.255.255.0
```

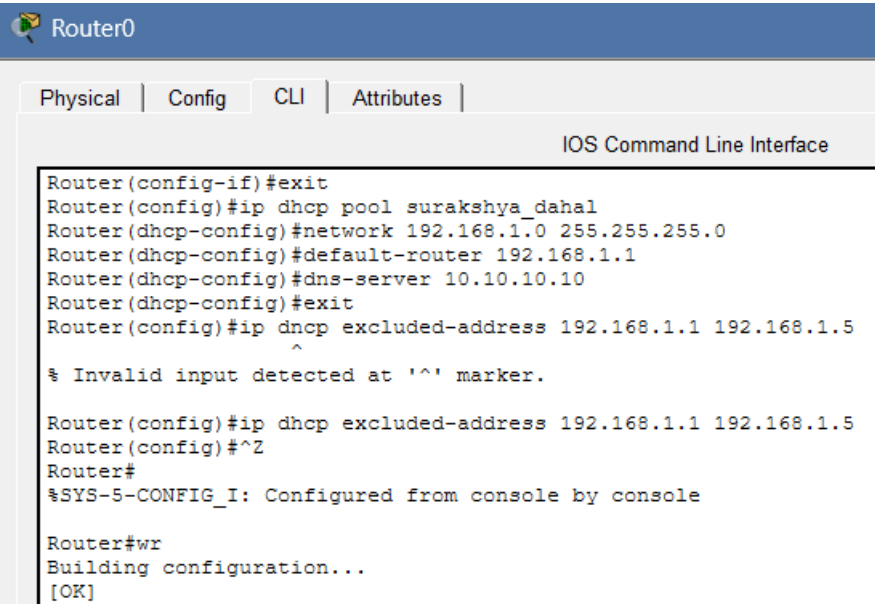
```
Router(dhcp-config)# default-router 192.168.1.1
```

```
Router(dhcp-config)# dns-server 10.10.10.10
```

5. Exit and save the configuration:

```
Router(dhcp-config)# exit
```

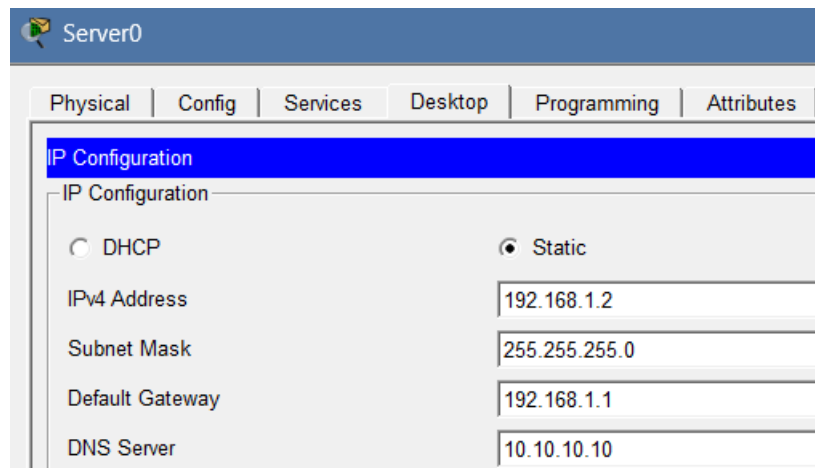
```
Router# wr
```



```
Router0
Physical | Config | CLI | Attributes |
IOS Command Line Interface
Router(config-if)#exit
Router(config)#ip dhcp pool surakshya_dahal
Router(dhcp-config)#network 192.168.1.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.1.1
Router(dhcp-config)#dns-server 10.10.10.10
Router(dhcp-config)#exit
Router(config)#ip dhcp excluded-address 192.168.1.1 192.168.1.5
^
% Invalid input detected at '^' marker.
Router(config)#ip dhcp excluded-address 192.168.1.1 192.168.1.5
Router(config)#^Z
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#wr
Building configuration...
[OK]
```

4. Configuring the DHCP Server

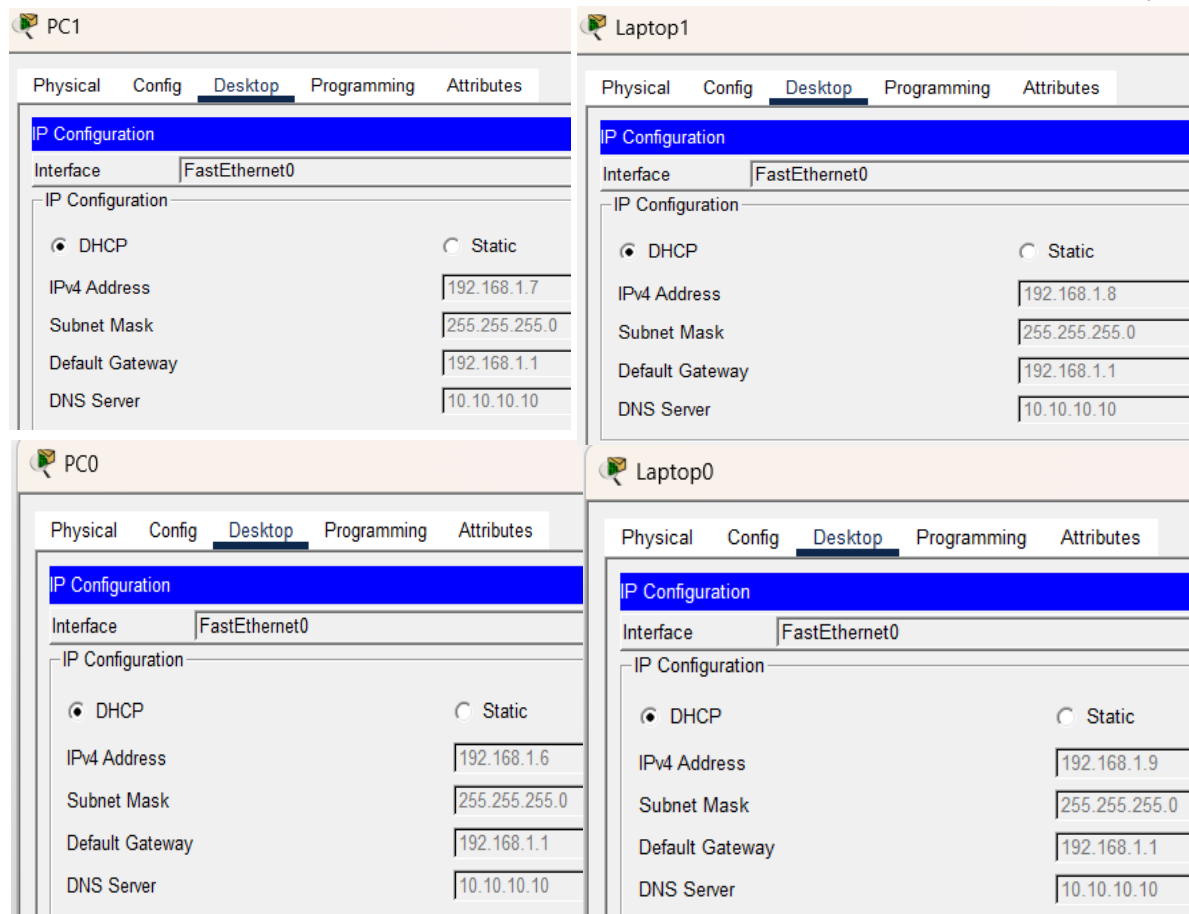
1. Open the Server.
2. Navigate to: Services → DHCP
3. Turn the DHCP service **ON**.
4. Create a DHCP pool.
5. Click **Add** to create the pool.



4. Configuring Clients

1. Open each PC and Laptop.
2. Navigate to: Desktop → IP Configuration
3. Select: DHCP
4. Wait for the device to obtain an IP address automatically.
5. Verify that valid IP addresses have been assigned.

6.



7. Testing Connectivity

1. Open Command Prompt on a client device (P0)
2. Ping the router: *ping 192.168.1.1*
3. Ping the server: *ping 192.168.1.2*
4. Verify successful replies.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time=14ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 14ms, Average = 3ms
```

Fig: Pinging the router (192.168.1.1) from

```
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

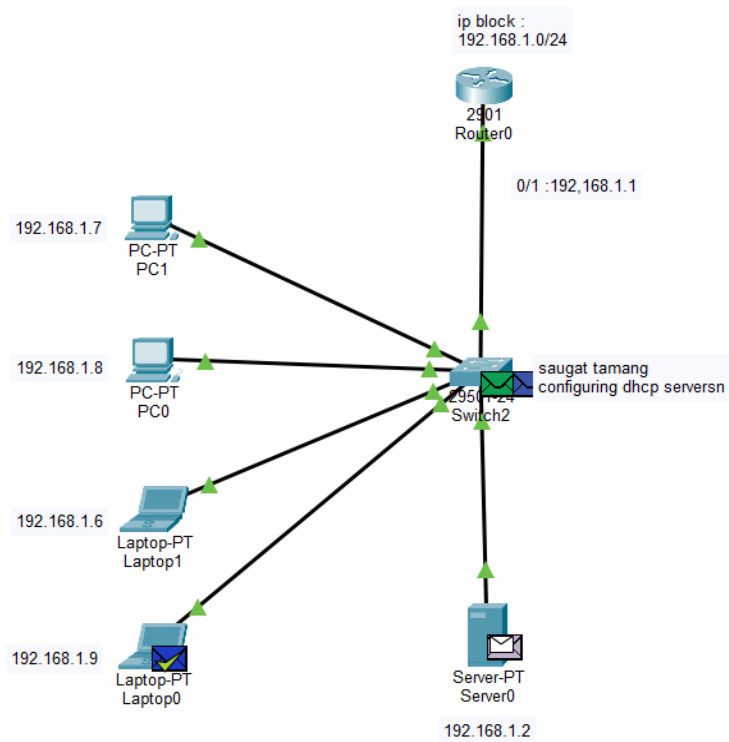
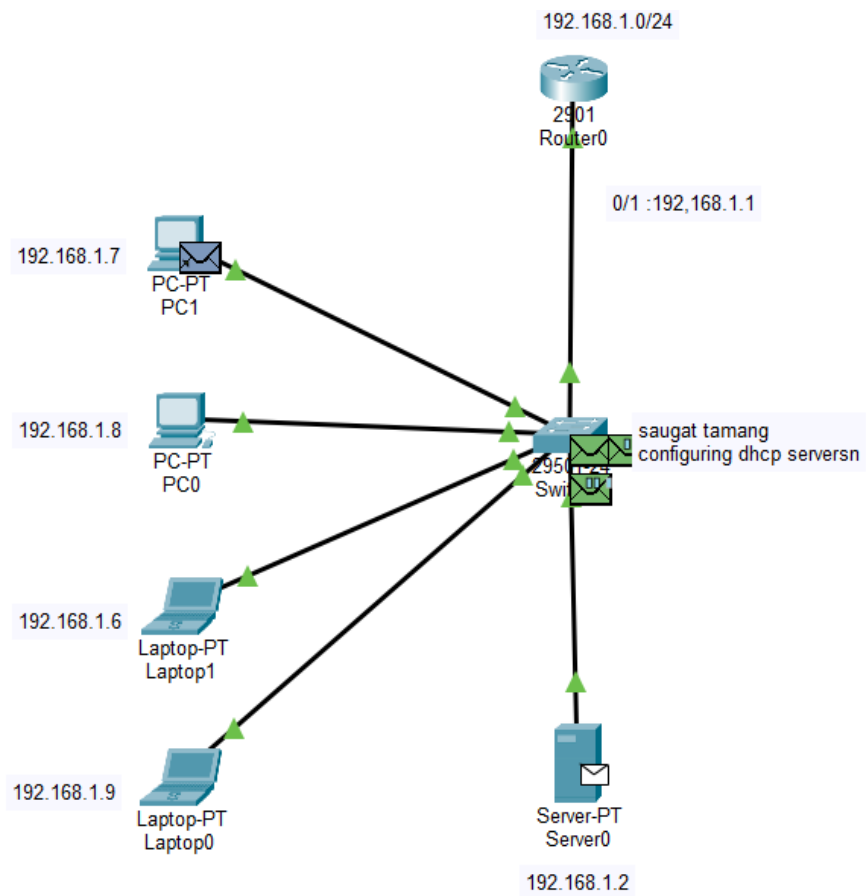
C:\>
```

Fig: Pinging the server (192.168.1.2) from client (pc 0,1 or laptop 0,1)

8. Simulation Verification

1. Switch Packet Tracer to **Simulation Mode**.
2. Generate network traffic using the **ping** command. (*pinging PC0 to Laptop1*)
3. Observe packet exchange between the DHCP server and clients.
4. Verify successful network communication.

Output



Conclusion

This lab successfully combined the study of Windows command-line diagnostics with automated network design. Utilizing commands like `ipconfig`, `ping`, `tracert`, `nslookup`, `netstat`, and `arp` provided essential hands-on experience in network monitoring and troubleshooting. Furthermore, by configuring a DHCP server in Cisco Packet Tracer, the lab demonstrated a shift from the manual addressing used in Lab 1 to an automated model. The seamless distribution of IP configurations and subsequent connectivity checks proved the scalability and efficiency of dynamic IP management.